consequences, abandoning the guess, and going on to the next guess, one guess after another after another after another, until exhaustion sets in. Then they begin again the next day.

A bit later in the movie, when the Professor is on his deathbed, he confesses to Murph: “I lied, Murph. I lied to you.” It is a poignant scene. Murph infers that he knew something was wrong with his equation, knew from the outset. And Dr. Mann tells the Professor’s daughter as much in an equally poignant scene on Mann’s planet.

But, in fact—Murph realizes, soon after the Professor’s death—“His solution was correct. He’d had it for years. It’s half the answer.” The other half can be found inside a black hole. In a black hole’s singularity.

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41 Gravity field and steady-state Ocean Circulation Explorer, GOCE.

42 According to Einstein’s relativistic laws the dark energy that (presumably) makes the expansion of our universe accelerate has a second effect: It produces an enormous tension in our brane, like the tension in a stretched rubber band or rubber sheet. And Einstein’s laws also dictate that, in order for spacetime outside the AdS sandwich to be free of warping, as we desire, each confining brane must have internal pressure that is half as big as our own brane’s internal tension. It is this pressure that is dangerous.

43 Or the buckling could make one or both branes spring outward, releasing the AdS layer and so destroying Newton’s inverse square law and making the planets all fly away from the Sun —not quite so bad for our universe, but pretty miserable for humans.

44 The meanings of the various symbols in the equation are spelled out on the Professor’s other fifteen blackboards, along with other information about the equation, all of which I ghost-wrote for the movie’s filming. You can see photographs of all sixteen blackboards on this book’s page at Interstellar.withgoogle.com.